Experiment 1

Signifigant figures and errors

expirement (Signifigant figures and Errors)

أسماء إسحاق محمد لغواصنة

$$a.\ average = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$average = \frac{4.32 + 4.35 + 4.31 + 4.36 + 4.37 + 4.34}{6}$$

average: 4.3416 (1 S.F)

average: 0.4×10^{1}

$$average\ deviation = \sqrt{\frac{\Sigma(\bar{x} - x_i)^2}{N-1}}$$

$\sigma = 10$

SMURFS POLYTECHNIC UNION
إتحاد سنافر
البوليتكنك
SPU

أ سماء سحاق محمد الفواضلة

c.

i) (3.42 + 0.04 = 3.46 and 3.48 (yes)
(3.42 − 0.04 = 3.38 and 3.48 (NO)

ii) (13.106 + 0.014 = 13.12 and 13.206 (NO)
(13.106 − 0.014 = 13.092 and 13.206 (NO)

iii) $(2.95 + 0.03)\times10^{8} = 2.98\times10^{8}$ and 3×10^{8} (yes)
$(2.95 - 0.03)\times10^{8} = 2.92\times10^{8}$ and 3×10^{8} (NO)

d:

i) $Z = 4.72 - (2.5 \times 4.4) + 15.63$
$Z = (4.72 - 11) + 15.63$
$Z = -6.2 + 15 + 63$
$Z = 9.4$ m

$\Delta Z = 0.12 - (2.5 \times 0.2) + 0.16$
$\Delta Z = -0.22$ m

ii) $Z = \frac{(14.42 \times 3.61)}{650}$
$Z = 0.080$ m/sec

$\Delta Z = \frac{(0.03 \times 0.18)}{20}$
$\Delta Z = 0.00027$ m/sec

iii) $Z = (3.55)^3$
$Z = 44.7$ m

$\Delta Z = (0.15)^3$
$\Delta Z = 0.0034$ m

iv) $Z = 0.644\,((3.42 \times 5.00) + 12.13)$
$Z = 0.644\,(17.1 + 12.13)$
$Z = 0.644\,(29.23)$
$Z = 18.8$ m^2

$\Delta Z = 0.004\,((0.06 \times 0.12) + 0.08)$
$\Delta Z = 0.004\,(0.0072 + 0.08)$
$\Delta Z = 0.004\,(0.0872)$
$\Delta Z = 0.0003$ m^2

e:

i)(0.000 42): (2 S.F)

ii)(0.147 00): (5 S.F)

iii)(4.2×10^{6}): (2 S.F)

iv)(-154.090×10^{-27}): (6 S.F)

أسماء إسحاق محمد الفواضلة

f:

$$\text{average Length} = \frac{123 + 123 + 124 + 123 + 123}{5}$$

average Length: 123.2 mm

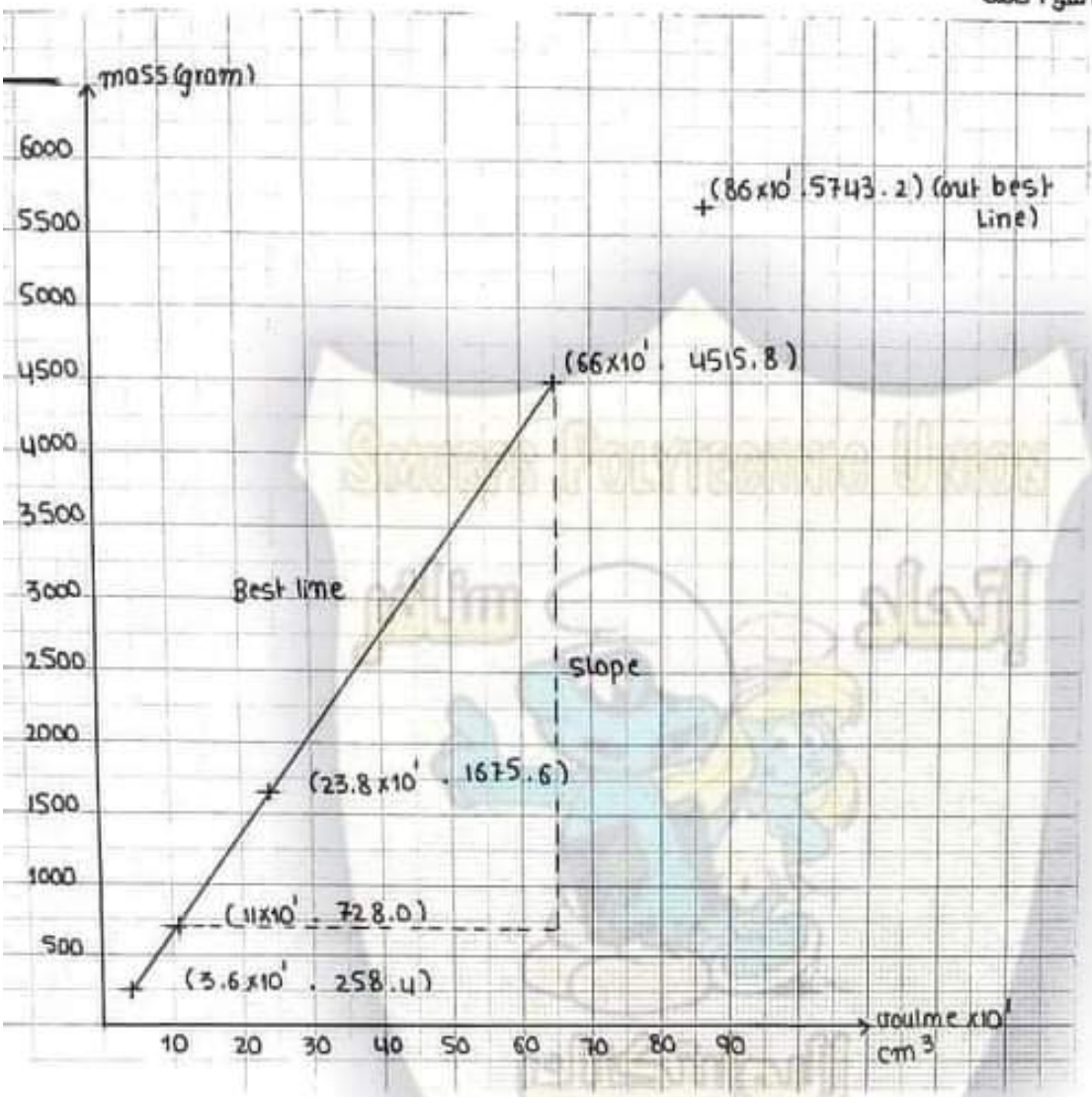
$$\text{uncertainty in length} = \frac{123}{2}$$

uncertainty in length: 61.5 mm

g:

Side length L(cm) ΔL: 0.05 cm	mass (g) Δm: 0.05 g	volume(cm^3) ΔV:	density: $\frac{m}{V}$ $d(g/cm^3)$	Δvolume $\Delta V(cm^3)$
3.3	258.4	3.6×10^{1}	7.2	0.05
4.8	728.0	11×10^{1}	6.6	0.03
6.2	1675.6	23.8×10^{1}	7.0	0.02
8.7	4515.8	66×10^{1}	6.8	0.02
9.5	5743.2	86×10^{1}	6.7	0.02

أسماء إسحاق محمد الفواضنة



$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{\Delta \text{mass}}{\Delta \text{volume}}$$

$$\text{Slope:} \frac{m_2 - m_1}{v_2 - v_1}$$

$$\text{Slope:} \frac{4515.8 - 728.0}{660 - 110} : 6.88 \ g/cm^3$$

Slope: density

density: $6.88 \ g/cm^3$

Experiment 2

Measurment

expire ment (mea sur ment)

We did not use the Significant figures rules

أسماء إسحاق محمد الفواضلة

value	Δ value
$m = 16.3$ gram	$\Delta m = 0.1$ gram
$d = 15.86$ mm	$\Delta d = 0.005$ mm
$r = 7.98$ mm	$\Delta r = 2.5 \times 10^{-3}$ mm
$A = 799.82$ mm^2	$\Delta A = 0.626 \times 10^{-3}$ mm^2
$V = 2074.34$ mm^3	$\Delta V = 0.939 \times 10^{-3}$ mm^3
$\rho = 7.85 \times 10^{-3}$ g/mm^3	$\Delta \rho = 2.15 \times 10^{-11}$ g/mm^3

For Sphere

value	Δ value
$m = 184.3$ gram	$\Delta m = 0.1$ grame
$L = 19$ mm	$\Delta L = 0.005$ mm
$d = 12.68$ mm	$\Delta d = 0.005$ mm
$r = 6.34$ mm	$\Delta r = 2.5 \times 10^{-3}$ mm
$A = 756.48$ mm^2	$\Delta A = 783.87 \times 10^{-7}$ mm^2
$V = 2398.06$ mm^3	$\Delta V = 4969.83 \times 10^{-7}$ mm^3
$\rho = 0.076$ g/mm^3	$\Delta \rho = 8.52 \times 10^{-12}$ g/mm^3

For Cylinder

value	Δ value
$m = 40.4$ gram	$\Delta m = 0.1$ gram
$L = 30.68$ mm	$\Delta L = 0.5$ mm
$w = 10$ mm	$\Delta Lw = 0.5$ mm
$h = 2.3$ mm	$\Delta h = 0.5$ mm
$A = 306.8$ mm^2	$\Delta A = 20.24$ mm^2
$V = 705.64$ mm^3	$\Delta V = 199.69$ mm^3
$\rho = 0.057$ g/mm^3	$\Delta \rho = 1.928 \times 10^{-4}$ g/mm^3

For Rectanguler

أسماء إسحاق محمد الفواضلة

for Sphere : الكرة

Area: $4\pi r^2$

volume: $\frac{4}{3}\pi r^3$

density: $\frac{\text{mass}}{\text{volume}}$

$\Delta A = 2\frac{\Delta r}{r}$

$\Delta v = 3\frac{\Delta r}{r}$

$\Delta \rho = (\frac{\Delta m}{m} + \frac{\Delta v}{v})\rho$

for Cylender : الأسطوانة

Area: $2\pi r L$

volume: $\pi r^2 L$

density: $\frac{\text{mass}}{\text{volume}}$

$\Delta A = (\frac{\Delta r}{r} + \frac{\Delta L}{L})A$

$\Delta v = (\frac{2\Delta r}{r} + \frac{\Delta L}{L})v$

$\Delta \rho = (\frac{\Delta m}{m} + \frac{\Delta v}{v})\rho$

for Rectangular : متوازي المستطيلات

Area: Lw

volume: Lwh

density: $\frac{\text{mass}}{\text{volume}}$

$\Delta A = (\frac{\Delta L}{L} + \frac{\Delta w}{w})A$

$\Delta v = (\frac{\Delta L}{L} + \frac{\Delta w}{w} + \frac{\Delta h}{h})v$

$\Delta \rho = (\frac{\Delta m}{m} + \frac{\Delta v}{v})\rho$

SMURFS POLYTECHNIC UNION
إتحاد سنافر
SPU

Experiment 3

Vectors



expiriment (vectors)

أسماء إسحاق محمد الفواضلة

	(1)	(2)	(3)
mass	0.235	0.19	0.19
θ(angle)	110	240	343
Force	2.35	1.9	1.9
F(cos(θ1)	-0.8037	-.95	1.817
F(sin(θ1)	2.208	-1.645	-0.55

$\Sigma F(x) = 0.064$

$\Sigma F(y) = 0.013$

Resultant $= \sqrt{(\Sigma F(x))^2 + (\Sigma F(y))^2} = 0.065$ Newton

θ(the angle = 15.71°

$|F3| = |R12| = \sqrt{F1^2 + F2^2 - 2F1\,F2\cos\theta 12}$ when $(\theta 12) = \theta 2 - \theta 1$

$R12 = 1.84$

$\theta(12) = 130$

$\cos(\theta 12) = -6.45$

$\beta = 50$

$$\frac{R12}{\sin\beta} = \frac{F1}{\sin\alpha}$$

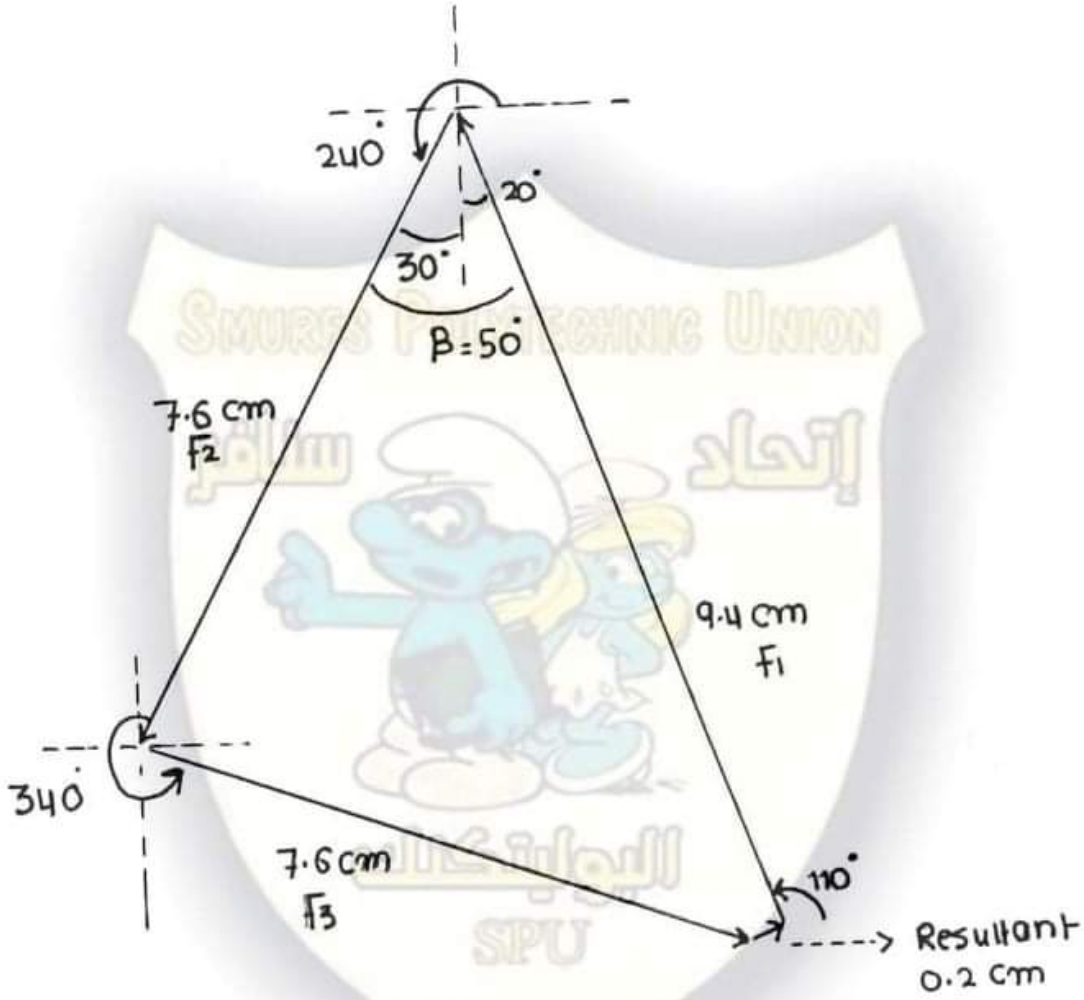
$$\frac{1.84}{\sin(50)} = \frac{2.35}{\sin\alpha}$$

$$\frac{1.66}{1.84} = \sin\alpha$$

$\alpha = 64.4°$

$\theta R12 = \theta 1 + \alpha$

$\theta R12 = 174.9$


240°
20°
30°
β=50°
7.6 cm
F2
9.4 cm
F1
340°
7.6 cm
F3
110°
Resultant
0.2 cm

Experiment 4

Newtens second law

expirement (Newtens Second Law)

case(1): when the total mass is constant

سماء إسحاق محمد الفواضلة

m(Kg)	t(sec)	F(Newton)	a(m/sec²)
0.01	1.69	0.1	0.351
0.02	1.19	0.2	0.70
0.03	0.975	0.3	1.05
0.04	0.844	0.4	1.40
0.05	0.755	0.5	1.75

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{\Delta F}{\Delta a} = \frac{F_2 - F_1}{a_2 - a_1}$$

$$\text{Slope} = \frac{0.3 - 0.1}{1.05 - 0.351}$$

Slope: 0.286

Slope = $m_w + m_h$ = (0.285 Kg)

case(2)

when the driving force (F(net)) is constant

m_w (Kg)	$m_w + m_h$	1/($m_w + m_h$)	t(sec)	a m/sec²
0.195	0.225	4.4	0.866	1.33
0.215	0.245	4.08	0.904	1.22
0.235	0.265	3.77	0.94	1.13
0.255	0.285	3.51	0.975	1.05
0.275	0.305	3.27	1.07	0.981

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{a_2 - a_1}{x_2 - x_1}$$

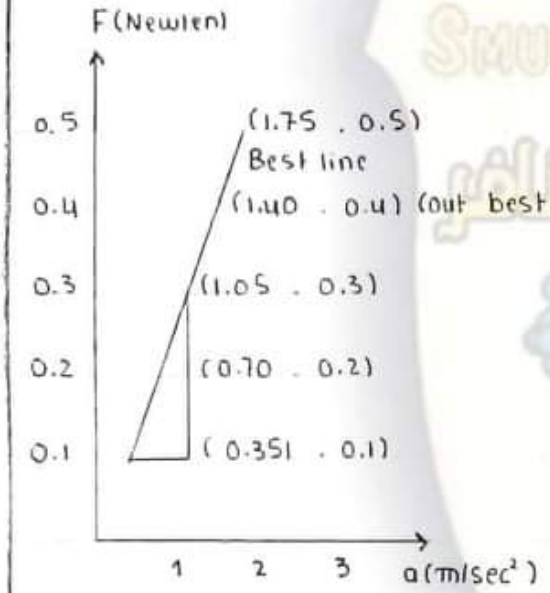
$$\text{Slope} = \frac{1.33 - 1.13}{4.4 - 3.77}$$

Slope: 0.3

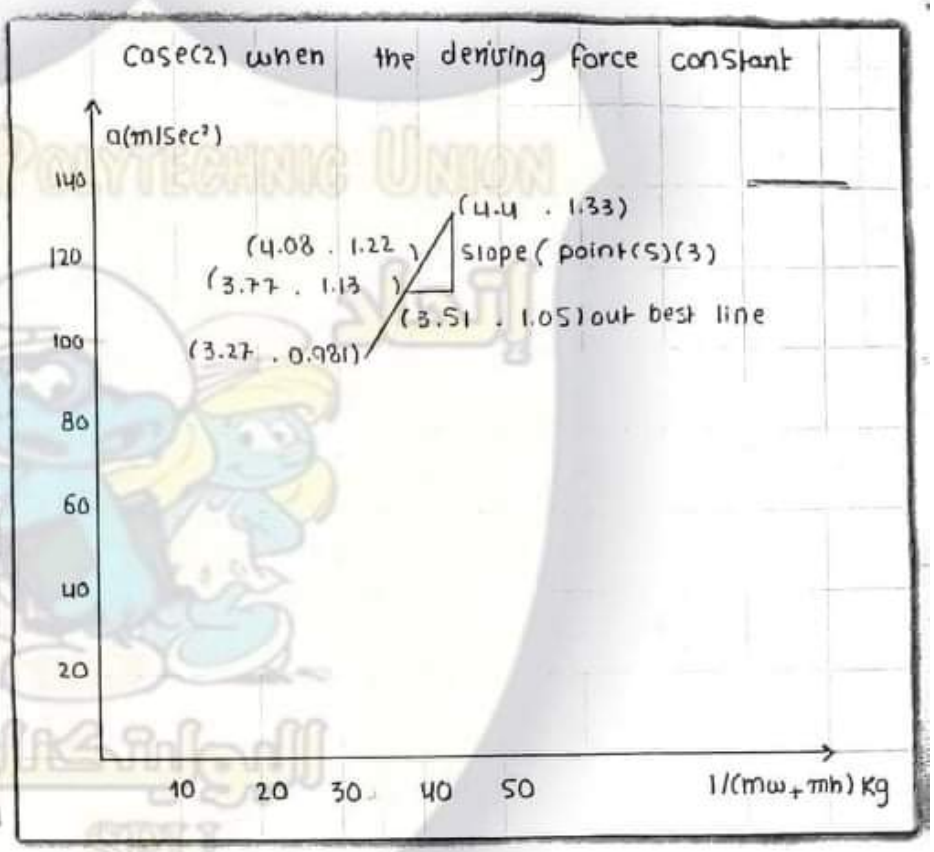
Slope: F(Net): 0.3 Newton

أسماء إسحاق محمد نوفل

case(1) when the total mass is constant



case(2) when the denising force constant



Experiment 5

Gravitational Potential Energy

أسماء إسحاق محمد لفواضلة

expirement(gravitational pote nial energy)

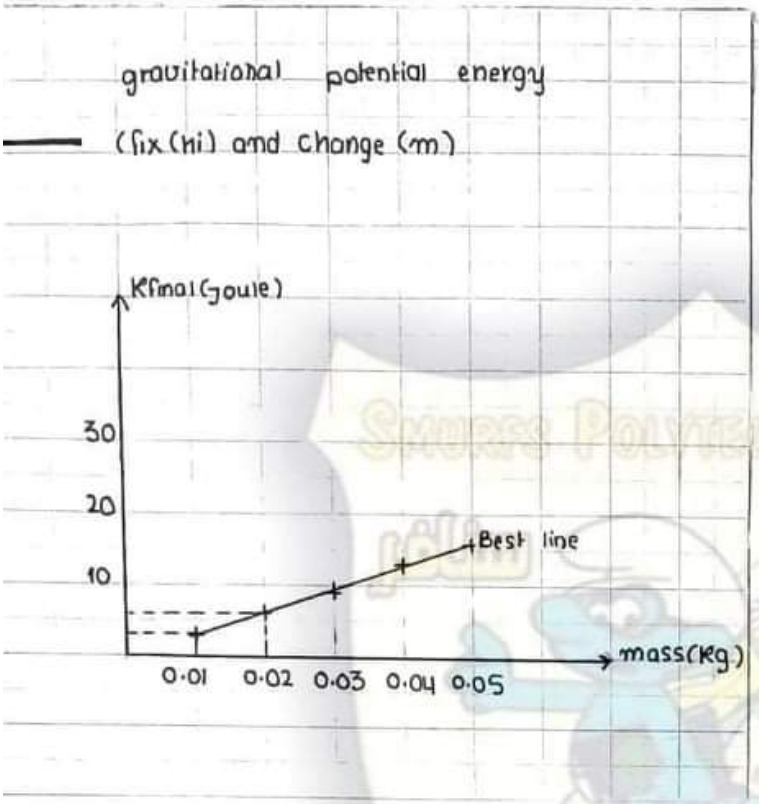
case(1)(fix (hi) and change (m)

L: 0.025 meter

M: 0.195 kg

hi(consiani): 0.3 meter

m(Kg)	t(sec)	$uf: \frac{L}{t}$	$(uf)^2$	$KF: \frac{1}{2}(m+M)uf^2$ (J)
0.01	0.0453	0.551	0.303	0.031
0.02	0.033	0.757	0.573	0.061
0.03	0.027	0.925	0.855	0.096
0.04	0.024	1.041	1.083	0.127
0.05	0.022	1.136	1.290	0.158



$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{\Delta KF}{\Delta m}$$

$$\text{Slope} = \frac{0.061 - 0.031}{0.02 - 0.01}$$

Slope: 3

Slope: $g\ hi$

$$g = \frac{\text{Slope}}{hi}$$

$$g = \frac{3}{0.3} = 10$$

g (gravity Constant): 10
(experimental value)

$$\text{persentage error} = \frac{|g_{true} - g_{exp}|}{g_{true}} \times 100\%$$

$$\text{perse ntage error} = \frac{|9.8 - 10|}{9.8} \times 100\%$$

persentage error: 2%

أسماء إسحاق محمد لفوانة

(gravalational potential energy)

أسماء إسحاق محمد الفواصلة

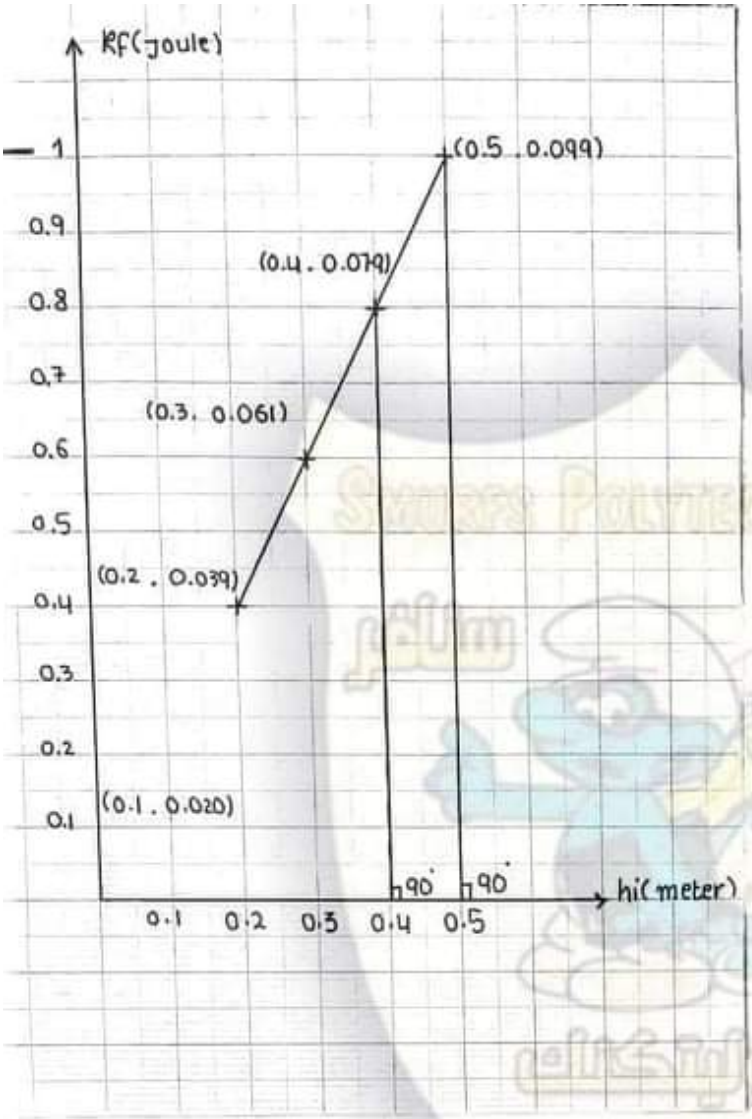
case(2) (fix (m) and change (hi))

L: 0.025 meter

m (constant) : 0.02 kg

M : 0.195 kg

hi(m)	t(sec)	$uf = \frac{L}{t}$	$(uf)^2$	$Kf = \frac{1}{2}(m+M)uf^2$
0.1	0.057	0.438	0.191	0.020
0.2	0.041	0.609	0.370	0.039
0.3	0.033	0.757	0.573	0.061
0.4	0.029	0.862	0.743	0.079
0.5	0.026	0.961	0.923	0.099



$$\text{Slope} : \frac{\Delta y}{\Delta x} = \frac{\Delta Kf}{\Delta hi}$$

$$\text{Slope} : \frac{0.099 - 0.079}{0.5 - 0.4}$$

$$\text{Slope} : 0.2$$

$$\text{Slope} : g\ m$$

$$g : \frac{\text{Slope}}{m}$$

$$g(\text{gravity constant}) = \frac{0.2}{0.02}$$

$$g = 10$$

(experemantal value)

$$\text{persentage error} = \frac{|g_{true} - g_{exp}|}{g_{true}} \times 100\ \%$$

$$\text{persentage error} = \frac{|9.8 - 10|}{9.8} \times 100\ \%$$

$$\text{persentage error} = 2\ \%$$

أسماء إسحاق محمد لغوا صنة

Experiment 6

One-Dimensional Collision

أسماء إسحاق محمد العواضلة

expire ment (One - Dimensional collision)

part (1): Elastic collision

1. Linear momentum before collision

$\vec{P_i} = m_1\vec{v}_{1i} + m_2\vec{v}_{2i}(-)$ when $\vec{v}_{1i} = \frac{L_1}{A_1}$, $\vec{v}_{2i} = \frac{L_2}{B_1}$

$\vec{P_i} = (0.2)\left(\frac{0.05}{0.2345}\right) + -(0.2)\left(\frac{0.025}{0.0250}\right)$

$\vec{P_i} = (0.2)(0.213) - (0.2)$ (direction to left)

$\vec{P_i} = -0.1574$ Kg. m /sec

2. Linear momentum after collision

$\vec{P_f} = -m_1\vec{v}_{1f} + m_2\vec{v}_{2f}$ when $\vec{v}_{1f} = \frac{L_1}{A_2}$, $\vec{v}_{2f} = \frac{L_2}{B_2}$

$\vec{P_f} = -(0.2)\left(\frac{0.05}{0.0500}\right) + (0.2)\left(\frac{0.025}{0.1174}\right)$

$\vec{P_f} = -(0.2) + (0.2)(0.213)$

$\vec{P_f} = -0.1574$ Kg. m/sec

Result ($\vec{P_i} = \vec{P_f}$) then the linear momentum in Elastic collision is Conserved

part (2) : Inelastic Collision

1. Linear momentum befor collision

$\vec{P_i} = -m_1\vec{v}_{1i} + m_2\vec{v}_{2i}$ (zero bec. const ant) when $\vec{v}_{1i} = \frac{L_1}{B_1}$

$\vec{P_i} = -(0.295)\left(\frac{0.025}{0.047}\right)$

$\vec{P_i} = -0.1566$ Kg. m/sec

2. Linear momentum after collision

$\vec{P_f} = (m_1 + m_2)(\vec{v}_f)(-)$ to the left when $\vec{v}_f = \frac{L_1}{A_2}$ OR $\frac{L_2}{A_1}$

$\vec{P_f} = -(0.295 + 0.2)\left(\frac{0.05}{0.1579}\right)$

$\vec{P_f} = -(0.495)(0.316)$

$\vec{P_f} = -0.156$ Kg. m /sec

Result ($\vec{P_i} = \vec{P_f}$) then the momentum in the Inelastic collision is Conserved

التعبير بالإشارات نتيجة تعويض إشارة (اتجاه السرعة) لأن الزخم هي الكمية متجهة

expire ment (One_Dimensional collision)

part(1): Elastic collision

1. Kinetic Energy before collision

$(KE)i = \frac{1}{2} m_1 \vec{v}_{1i}^{\,2} + \frac{1}{2} m_2 \vec{v}_{2i}^{\,2}$

$(KE)i = (0.5)(0.2)(0.213)^2 + (0.5)(0.2)(1)^2$

$(KE)i = (0.0045) + (0.1)$

$(KE)i = 0.1045$ joie

2. Kinetic Energy after collision

$(KE)f = \frac{1}{2} m_1 \vec{v}_{1f}^{\,2} + \frac{1}{2} m_2 \vec{v}_{2f}$

$(KE)f = (0.5)(0.2)(1)^2 + (0.5)(0.2)(0.213)^2$

$(KE)f = (0.1) + (0.0045)$

$(KE)f = 0.1045$ joie

Result $(KE)i = (KE)f$ then the Kinetic Energy in the Elastic Collision is Conserved

part(2): InElastic Collision

1. Kinetic Energy before collision

$(KE)i = \frac{1}{2} m_1 \vec{v}_{1i}^{\,2} + \frac{1}{2} m_2 \vec{v}_{2i}^{\,2}$ (zero)(constant)

$(KE)i = (0.5)(0.295)(0.532)^2$

$(KE)i = (0.1475)(0.2830)$

$(KE)i = 0.0417$ joie

2. Kinetic Energy after collision

$(KE)f = \frac{1}{2} (m_1 + m_2) \vec{v}_f^{\,2}$

$(KE)f = (0.5)(0.295 + 0.2)(0.316)^2$

$(KE)f = (0.2475)(0.0998)$

$(KE)f = 0.0247$ joule

Result $(KE)i \neq (KE)f$ then the Kinetic Energy in the InElastic collision is not conserved

Experiment 7

Uniform Circuler Motion

expirement (Uniform circuler motion) (case(1))

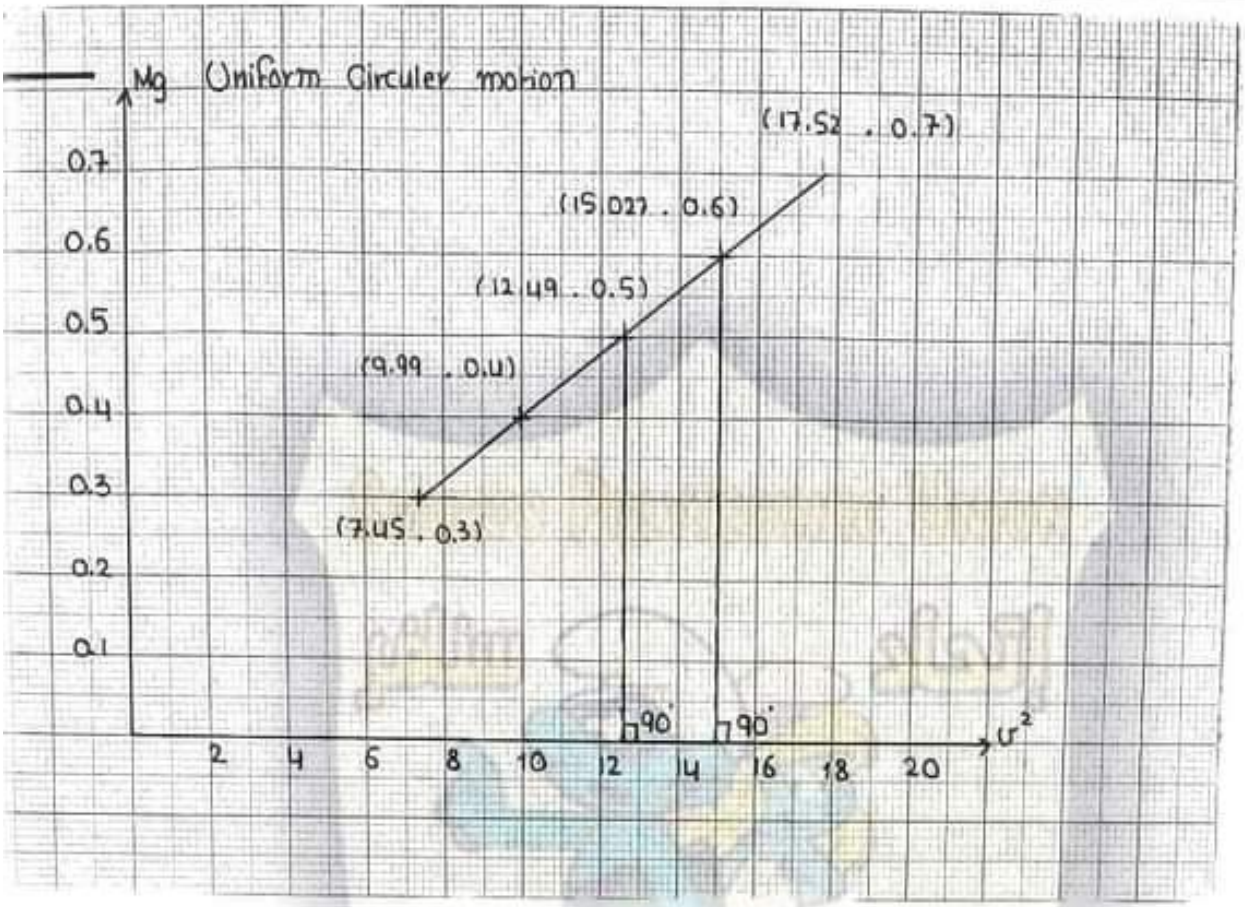
r = 0.3 m

m = 0.012 Kg

أسماء اسحاق محمد الفوا ضلة

المفروض نطبع بجواب تحت نفسه بالزبط

Hanging mass M(Kg)	Mg : Fc	time for 15 rev	period	Speed(v) (m/s)	$v^2 (m^2/s^2)$
0.03	0.3	10.32	0.688	2.73	7.45
0.04	0.4	8.94	0.596	3.16	9.99
0.05	0.5	8.00	0.533	3.534	12.49
0.06	0.6	7.29	0.486	3.876	15.027
0.07	0.7	6.75	0.45	4.186	17.52



Slope : $\frac{\Delta y}{\Delta x} = \frac{\Delta Mg}{\Delta v^2}$

Slope: $\frac{0.6 - 0.5}{15.027 - 12.49}$

أسماء إسحاق محمد لفواضلة

Slope: 0.039

Slope: $\frac{m}{r}$

m: Slope (r)

m: (0.039)(0.3)

m: 0.0117 = 0.012 kg

من التسمم

m(Kg from question: 0.012)

expirement (Uniform Curculer motion (case(2))

M(Hanging mass): 0.05 kg

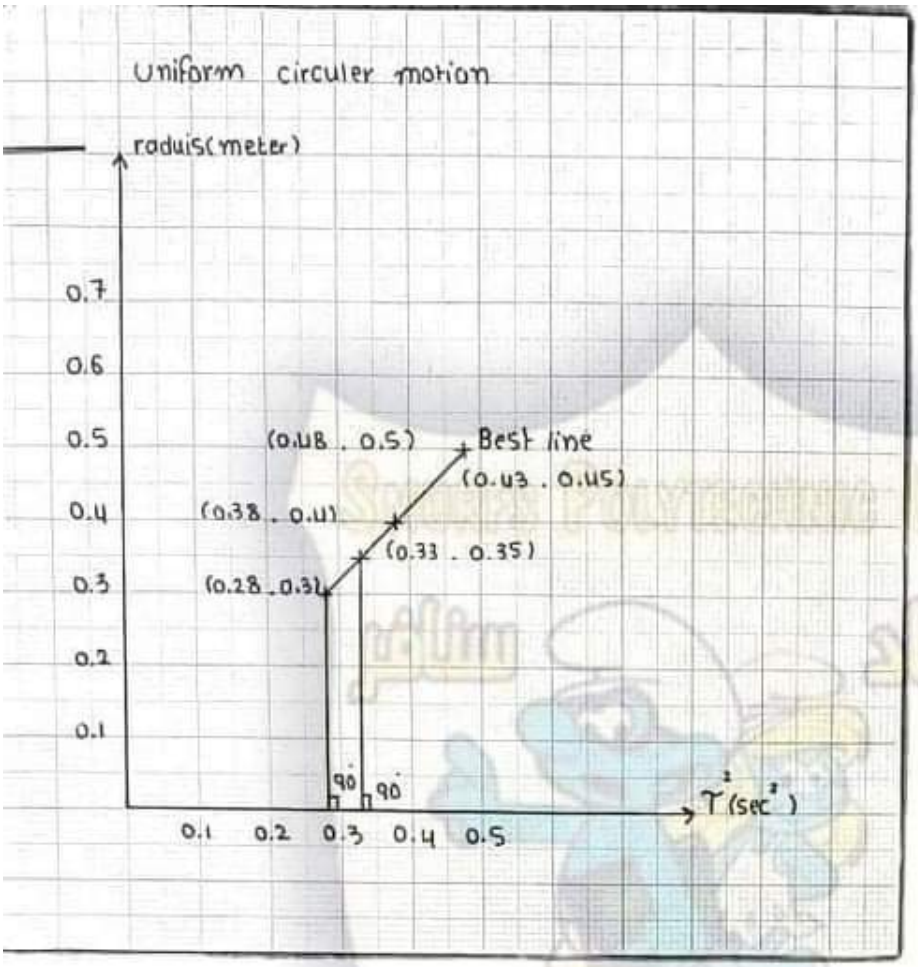
Mg = 0.5

m= 0.012 Kg

أسماء إسحاق محمد لفواضلة

المفروض يطلع الجواب تحت نفسه بالزبط

Raduis (m)	time for 15 rev	period (sec)	T^2 (Sec^2)	v^2 (m^2/sec^2)
0.3	8.00	0.533	0.28	12.49
0.35	8.64	0.576	0.33	14.56
0.4	9.2	0.613	0.38	16.79
0.45	9.80	0.653	0.43	18.72
0.5	10.32	0.688	0.48	20.82



$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{\Delta r}{\Delta T^2}$$

$$\text{Slope} = \frac{0.35 - 0.3}{0.33 - 0.28}$$

Slope = 1

$$m = \frac{Mg}{4(\pi)^2 (\text{slope})}$$

$$m = \frac{0.5}{4(3.14)^2 (1)}$$

$m = 0.012$ kg

أسماء إسحاق محمد لفواصلة